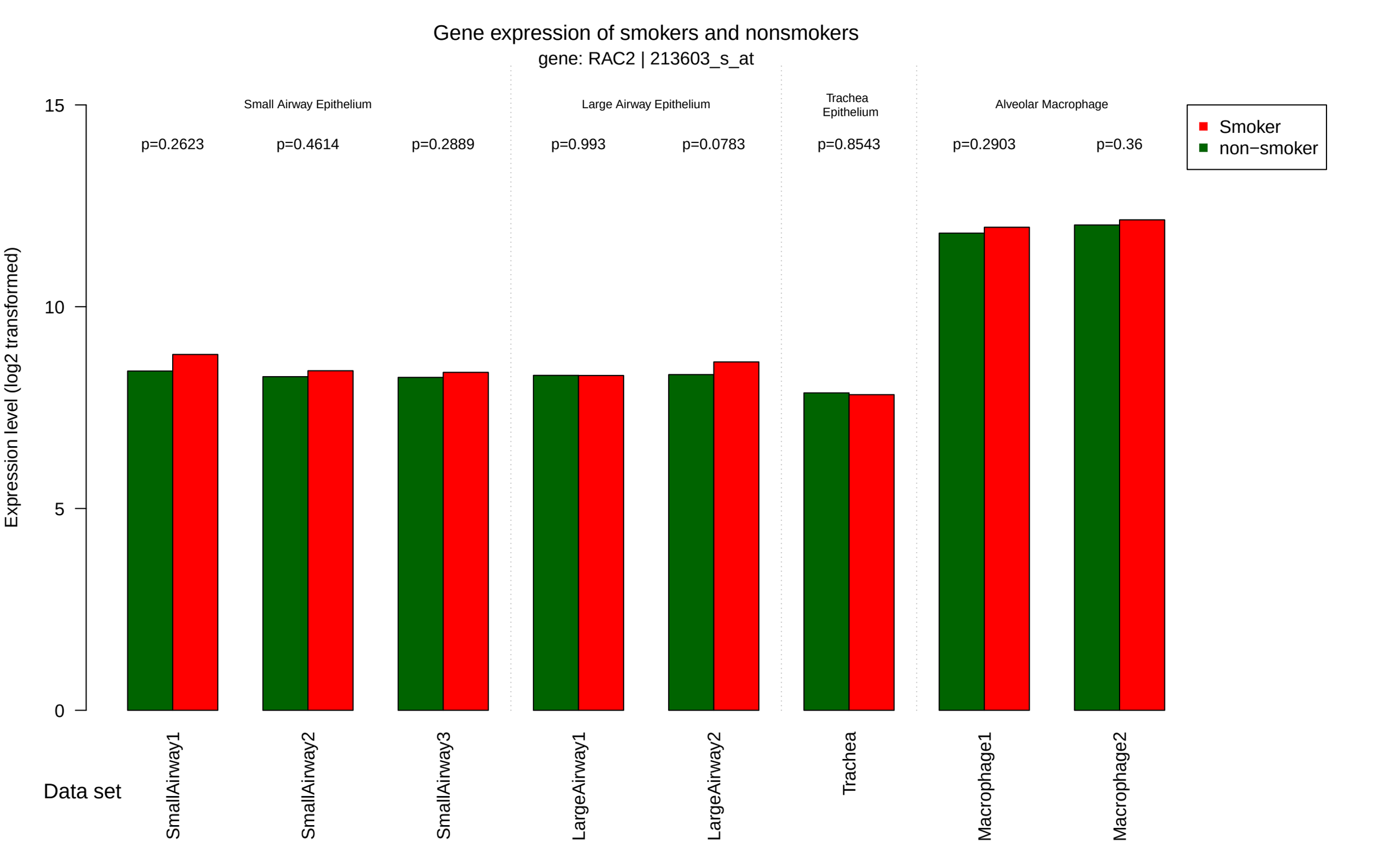
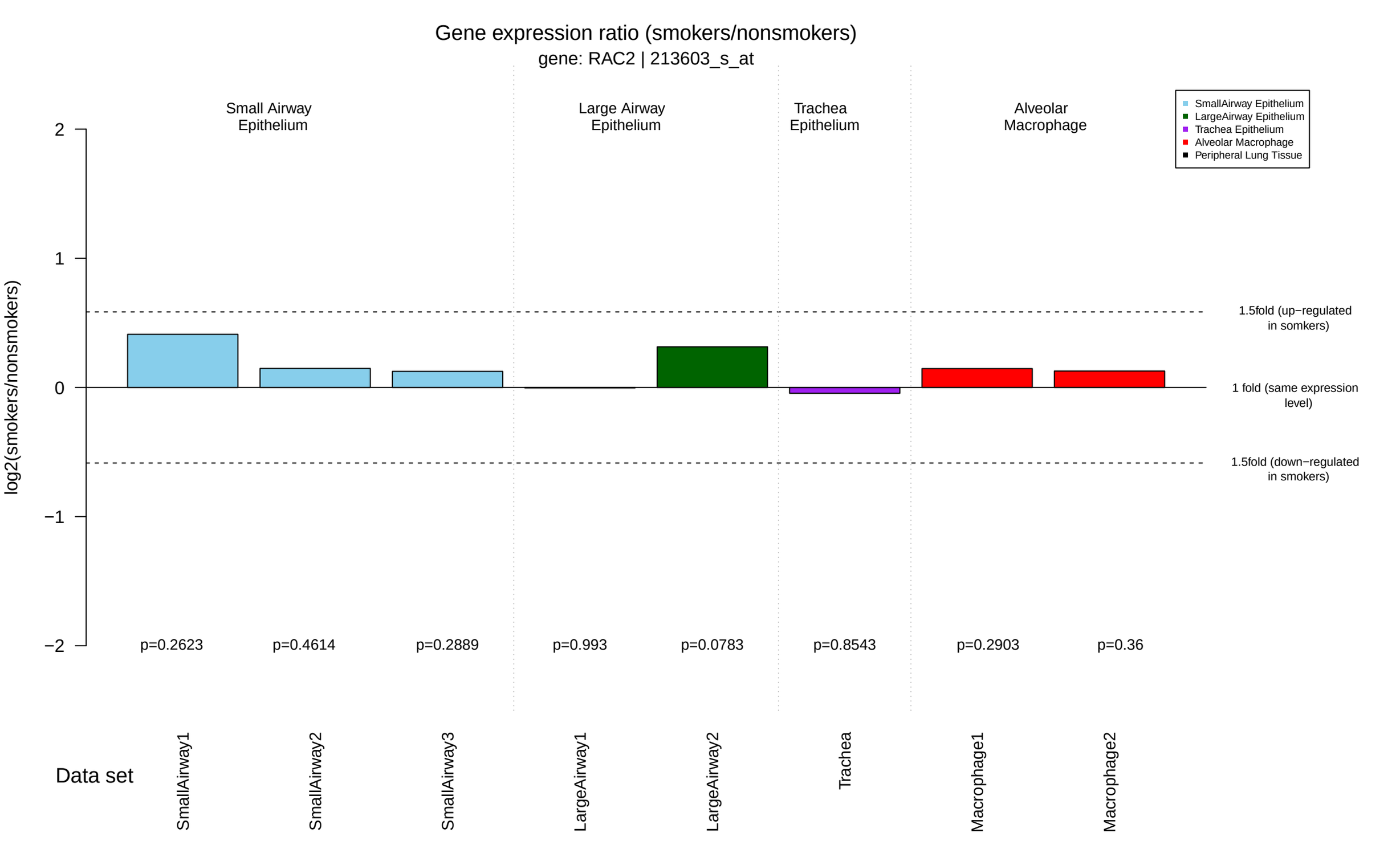
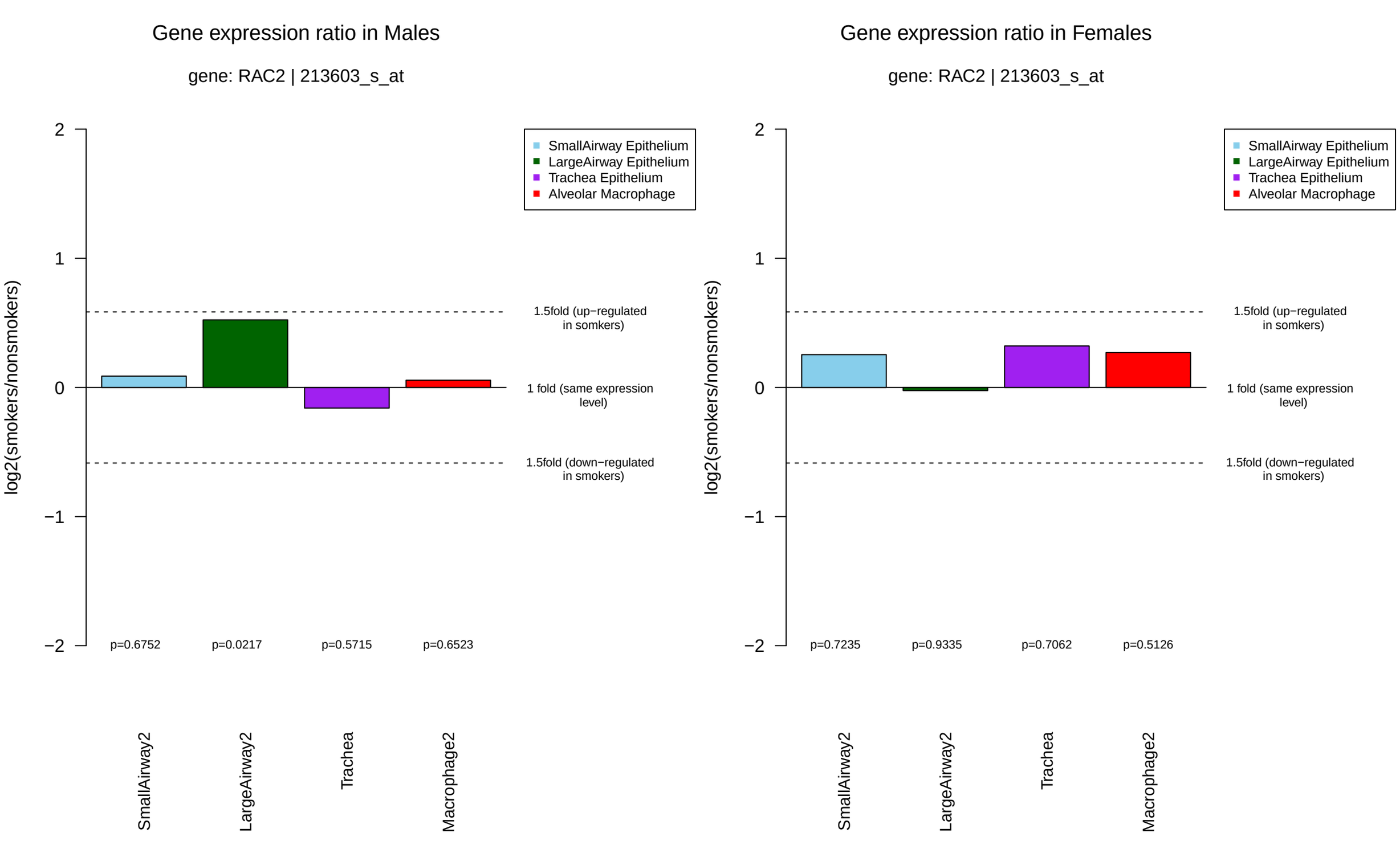
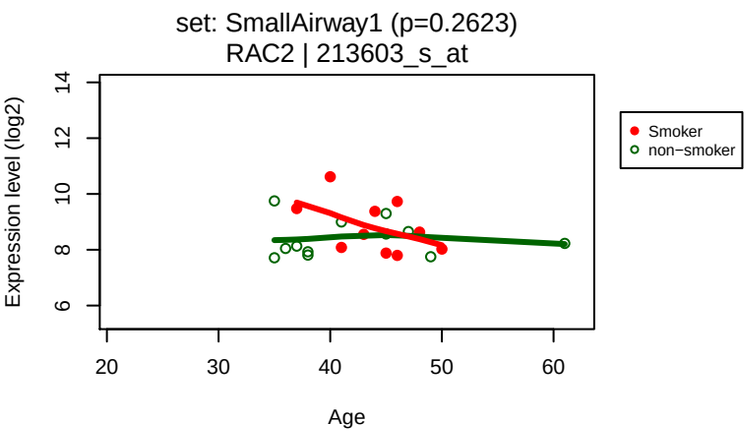
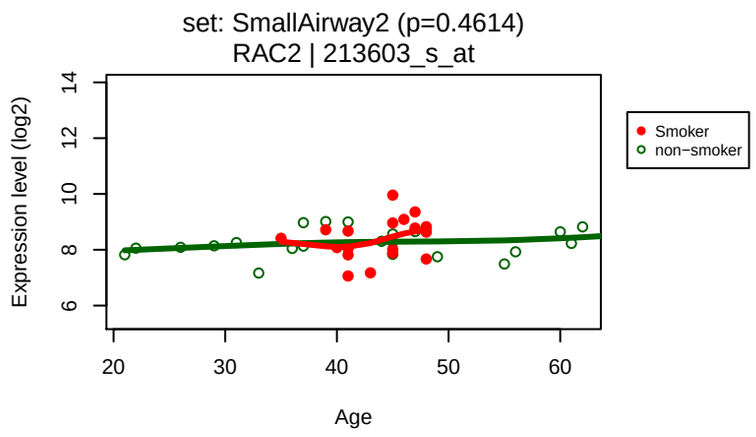
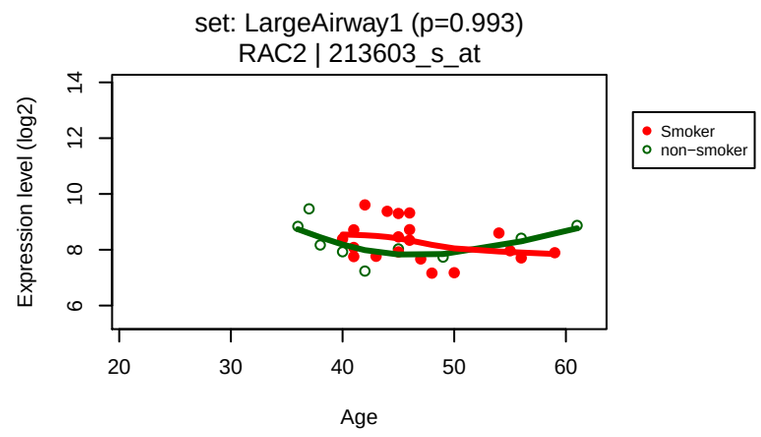

Gene expression of smokers and nonsmokers
gene: RAC2 | 213603_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2623
p=0.4614
p=0.2889
p=0.993
p=0.0783
p=0.8543
p=0.2903
p=0.36
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

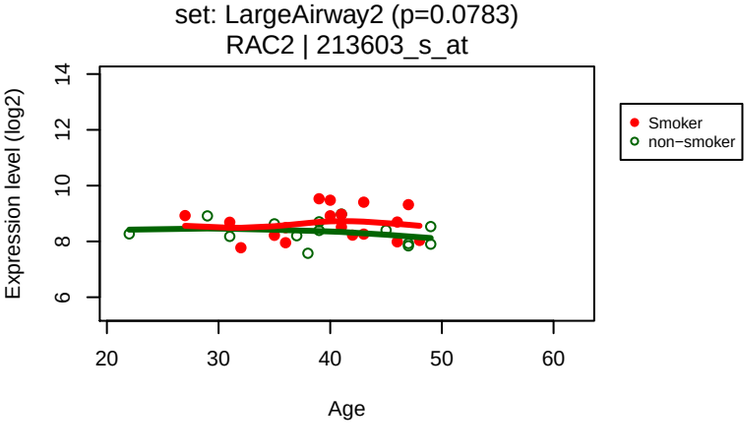

Gene expression ratio (smokers/nonsmokers)
gene: RAC2 | 213603_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2623
p=0.4614
p=0.2889
p=0.993
p=0.0783
p=0.8543
p=0.2903
p=0.36
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

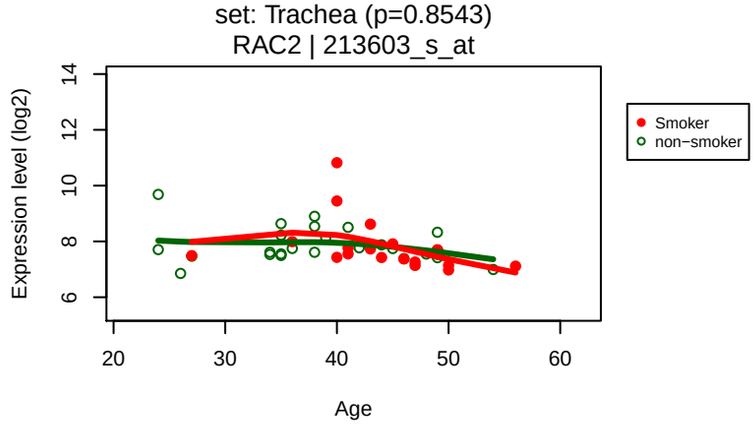

Gene expression ratio in Males
gene: RAC2 | 213603_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6752
p=0.0217
p=0.5715
p=0.6523
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAC2 | 213603_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7235
p=0.9335
p=0.7062
p=0.5126
SmallAirway2
LargeAirway2
Trachea
Macrophage2

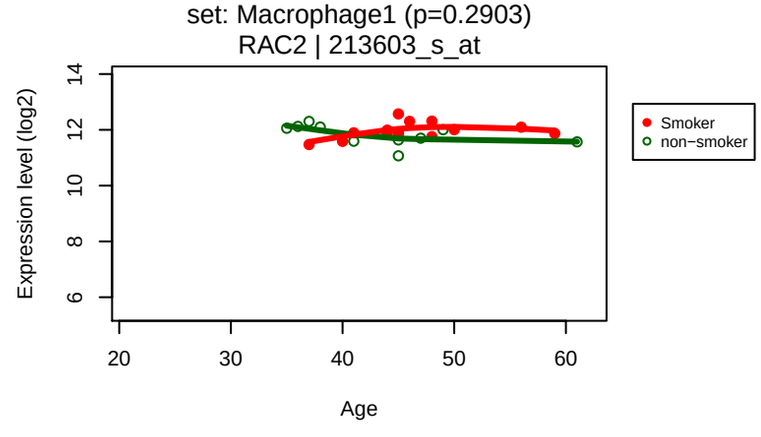

set: SmallAirway1 (p=0.2623)
RAC2 | 213603_s_at
Expression level (log2)
Age
Smoker
non−smoker

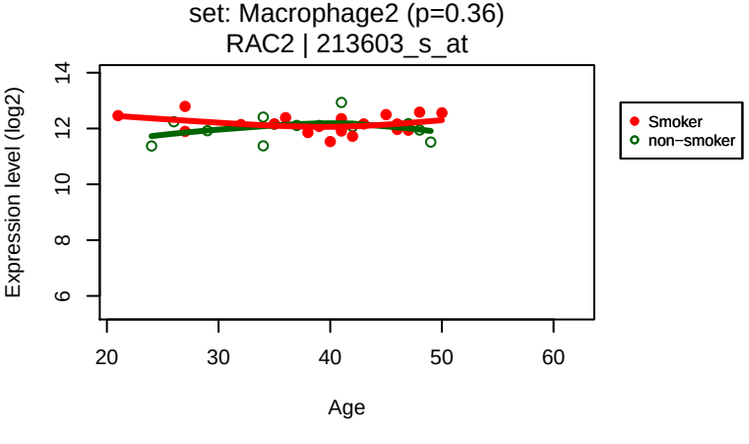

set: SmallAirway2 (p=0.4614)
RAC2 | 213603_s_at
Expression level (log2)
Age
Smoker
non−smoker

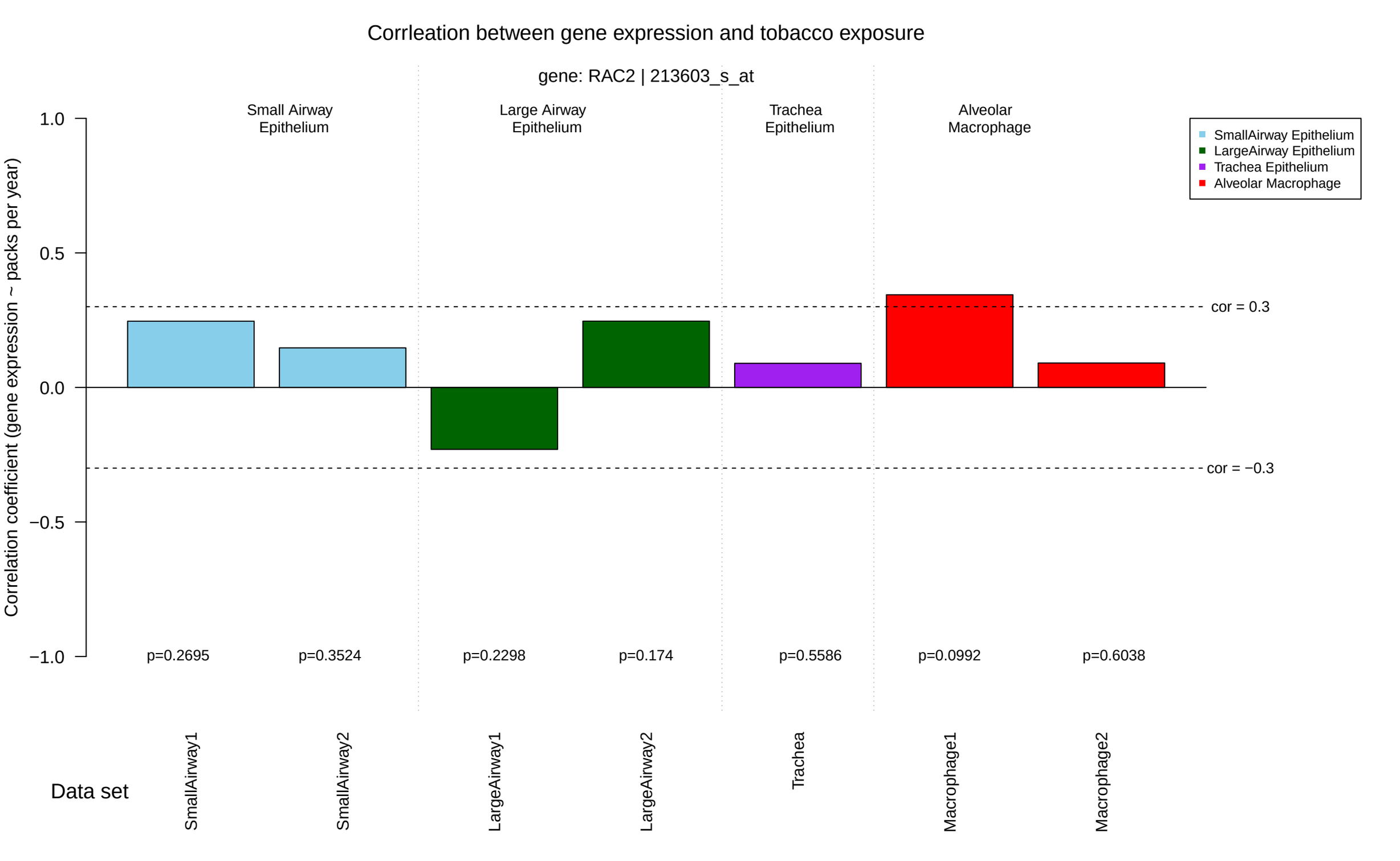

set: LargeAirway1 (p=0.993)
RAC2 | 213603_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0783)
RAC2 | 213603_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8543)
RAC2 | 213603_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2903)
RAC2 | 213603_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.36)
RAC2 | 213603_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAC2 | 213603_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2695
p=0.3524
p=0.2298
p=0.174
p=0.5586
p=0.0992
p=0.6038
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2